

# VP150 Final Recitation Class

## Moment of Inertia

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**1** Rigid Body

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# Rigid Body

**Definition:** The distance between any two points of the rigid body remains constant (no compression, stretch, etc.).

## Angular Quantities:

- angular displacement  $\theta$
- angular velocity  $\omega$
- angular acceleration  $\varepsilon$

## Comments:

- Note that  $\theta$ ,  $\omega$ , and  $\varepsilon$  are vectors.

$$d\vec{r} = d\vec{\theta} \times \vec{r}$$

- Compare them with  $x$ ,  $v$ , and  $a$ .

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# Moment of Inertia

## Definition:

$$I = \int r_{\perp}^2 dm$$

## Relationship with Kinetic Energy:

$$K = \frac{1}{2} I \omega^2$$

## Comments:

- $I$  is the moment of inertia about the fixed axis of rotation  $A$ .
- Moment of inertia depends on the distribution of mass.

# Moment of Inertia

## General Calculation:

An object rotate along z-axis with the density function  $\rho(x, y, z)$

$$I = \iiint_V (x^2 + y^2) \cdot \rho(x, y, z) \, dx \, dy \, dz$$

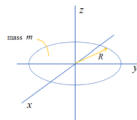
## Practical Calculation Steps:

- Determine the axis. Find out how the body is symmetrical. Construct the equation based on the symmetry.
- Find out the  $dm$ . Do integration.
- Substitute  $dm$  with  $m$  and other quantities.

# Moment of Inertia for Common Rigid Bodies

## Perpendicular axis of symmetry

$$I_A = \frac{1}{2}mR^2$$



$\sigma = \frac{m}{\pi R^2}$  — surface density of mass

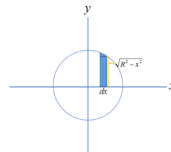
$$I_A = \int_{\text{object}} r_{\perp}^2 dm$$



$$dm = 2\pi r_{\perp} \sigma dr_{\perp}$$

## Axis of symmetry contained in the plane

$$I_B = \frac{1}{4}mR^2$$

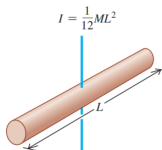




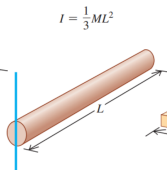
# Moment of Inertia for Common Rigid Bodies

**TABLE 9.2** Moments of Inertia of Various Bodies

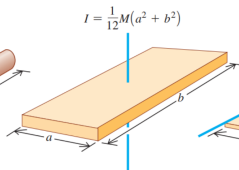
(a) Slender rod,  
axis through center



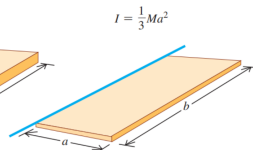
(b) Slender rod,  
axis through one end



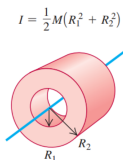
(c) Rectangular plate,  
axis through center



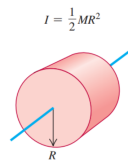
(d) Thin rectangular plate,  
axis along edge



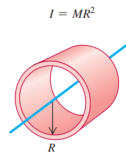
(e) Hollow cylinder



(f) Solid cylinder



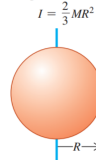
(g) Thin-walled hollow  
cylinder



(h) Solid sphere



(i) Thin-walled hollow  
sphere



# Theories

## Parallel Axis Theorem

$$I_{A'} = I_A + mb^2$$

\* $I_A$  is the Mol with the axis passing through the center of mass.

## Perpendicular Axis Theorem

$$I_Z = I_X + I_Y$$

\*The rigid body is only on the plane of  $XoY$ .

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## Ex.1

Two thin, uniform rods with mass  $m$  and length  $l$  are symmetrically connected to form a T-square ruler. Each part of the ruler is provided with a rotating shaft perpendicular to the ruler plane and the moment of inertia is  $I$ . Find  $I_{min}$  and  $I_{max}$ .

## Ex.2

A uniform square thin plate has a mass of  $m$  and each side length of  $a$ . Take the rotating axis through center  $O$  on the plane of the plate, and find the moment of inertia of the plate with respect to the axis.

## Ex.3

The semi-major axis length of an elliptical ring is  $a$ , the semi-minor axis length is  $b$ , and the mass is  $m$  (not necessarily homogeneous). The moment of inertia about the major axis is  $I_a$ , find the moment of inertia  $I_b$  about the minor axis.